Satellite Communication

Introduction to Satellite Subsystems

Introduction to Satellite Subsystems

In satellite communication system, various operations take place, among which the main operations are orbit controlling, altitude of satellite, monitoring and controlling of other subsystems

Earth Segment

- It presents in the ground segment which have the ability to access the satellite repeater in order to provide the communication between the users
- It is also called as ground segment

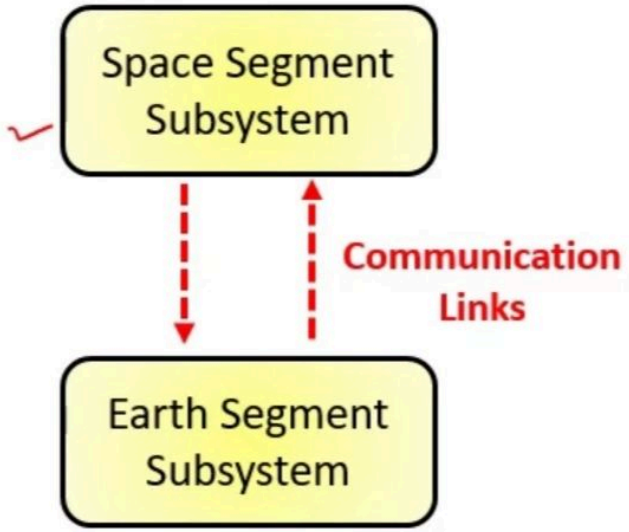
Space Segment

It is present in space, major space segment subsystems are:

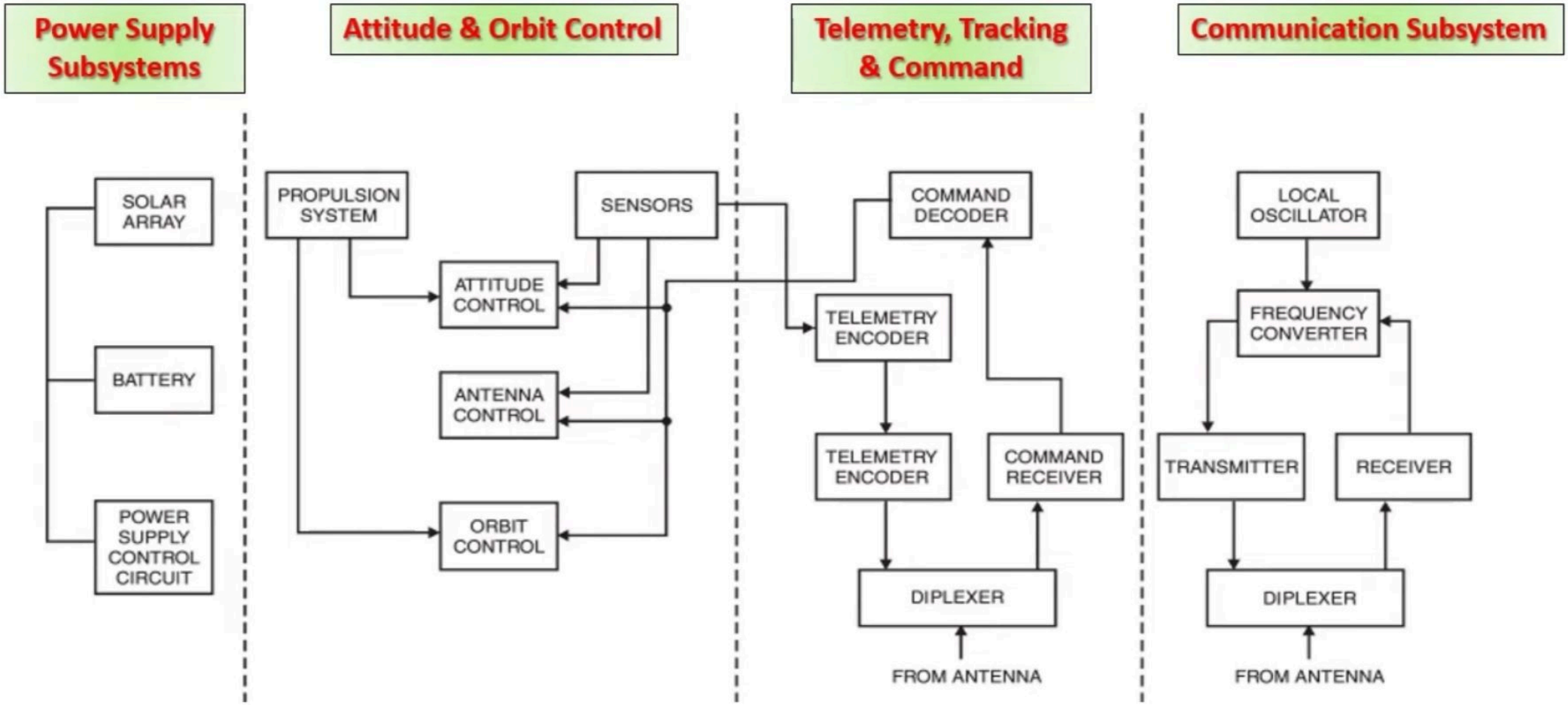
- AOC Subsystem
- TT&C Subsystem
- Power and Antenna Subsystems
- Transponders

A satellite communication consists of mainly **two segments**:

- Space Segment
- Earth Segment



Introduction to Satellite Subsystems



Satellite Communication

Introduction to Global Positioning System (GPS)

Introduction to GPS

GPS Development

- **Scientist John Hopkins** has developed a technique of using radio signals from Satellite, which provides **accurate position** updates to **navigation equipment** of **US Navy's ship** and **Submarines** (in late 1950's)
- In year 1973, Navy and Air Force program combined and formed **Navigational Technology Program** later which is evolved into **Navstar Global Positioning System (GPS)**
- Navstar **GPS** became the **world first position, navigation and timing information service** which was later offered to civilian users around the globe

Introduction to GPS

GPS Basics

- Global Positioning System **(GPS)** is a **navigation system** based on satellite
- It has created the revolution in navigation and position location
- It is mainly **used** in positioning, navigation, monitoring and surveying applications

- The major **advantages** of satellite navigation are real time positioning and timing synchronization
- Therefore **Satellite navigation systems** have become an integral part in most of the applications, where mobility is the key parameter

Introduction to GPS

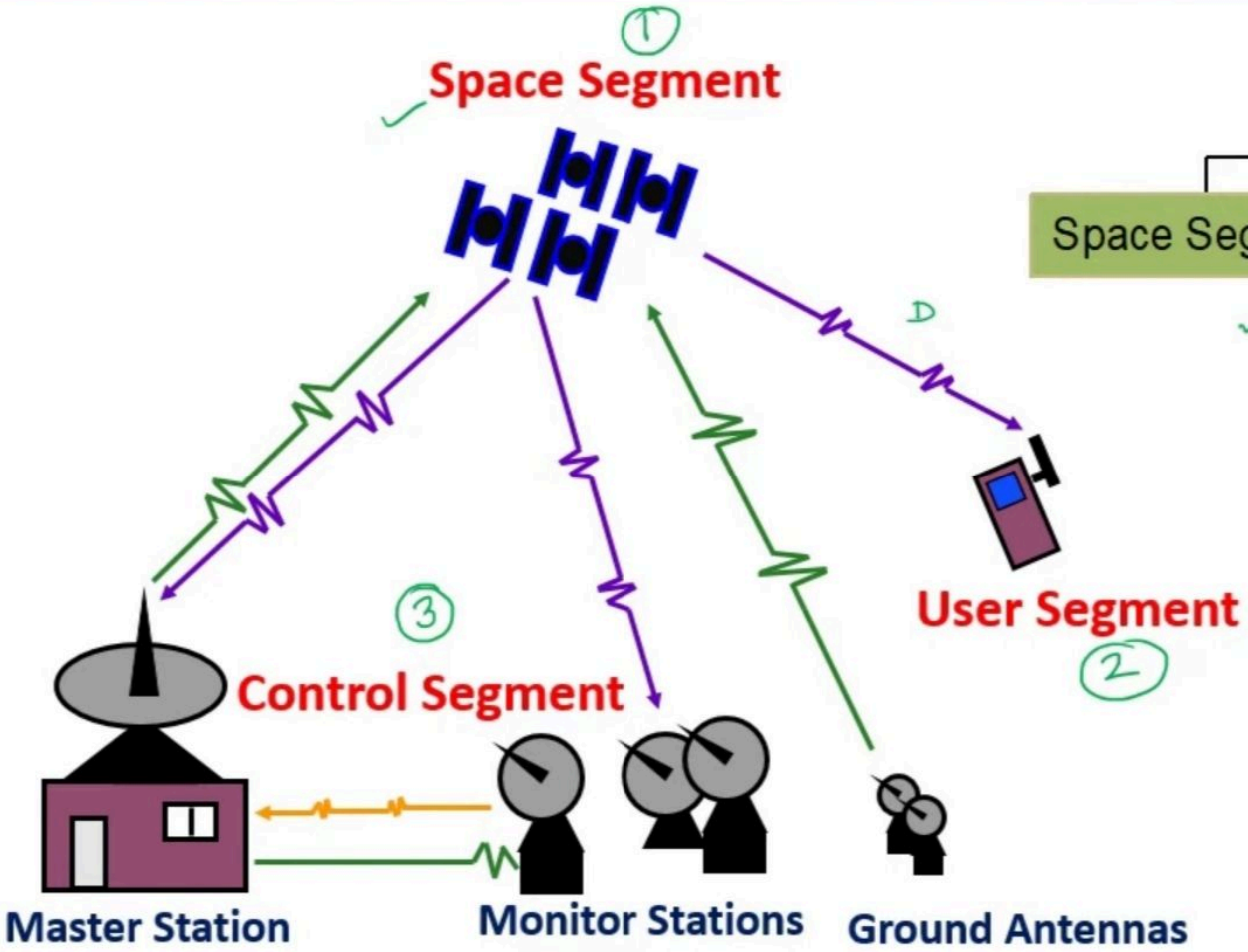
GPS Technical Structure

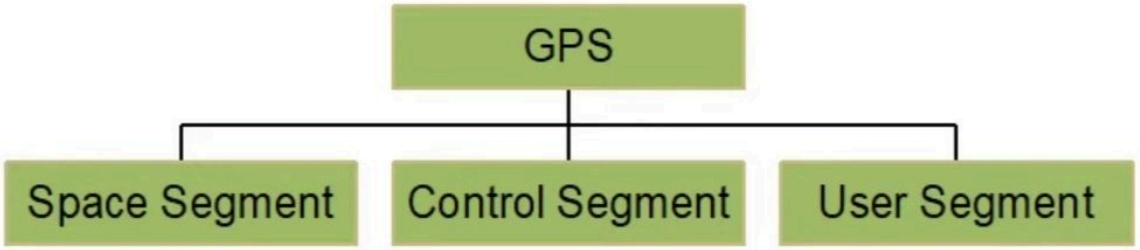
- The **GPS Space segment** consist of 24 Satellites orbiting in MEO
- Satellite Orbits are at an **altitude** of 20,200 Km with an **Orbital Inclination** of 55^0
- The Satellites are **clustered** in a group of **four**
- **Orbital Period** is about **one half a sidereal day** i.e. 11 hrs 58 Minutes
- Large number of GPS receivers can operate simultaneously
- **Example** of other position location principle, **LORAN (Long Range Navigation)** but accuracy and reliability is less

GPS provides:

- 24 hours service world wide
- Accurate and 3 dimensional location information (longitude, latitude & altitude)
- Provides continuous real time information
- Service to unlimited number of world wise users

Components of GPS

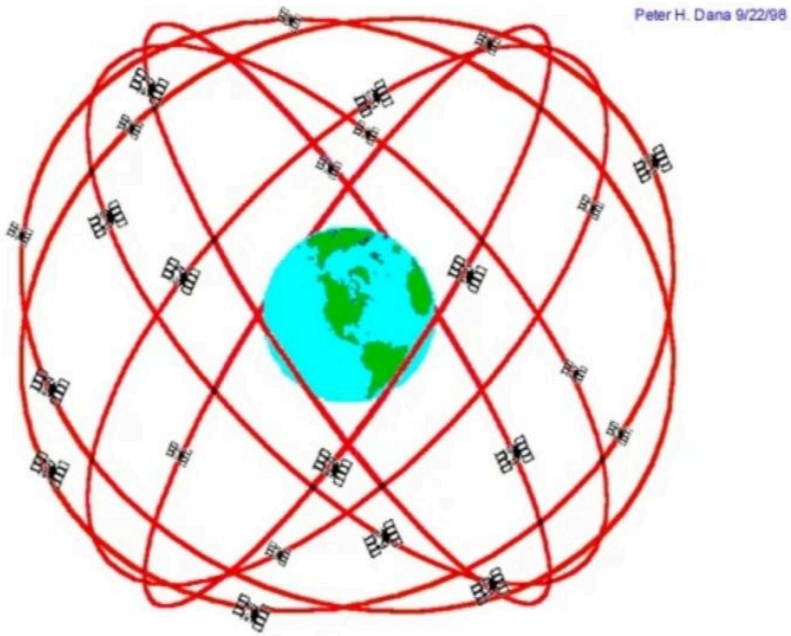




Introduction to GPS

Space Segment

- 24 GPS space vehicles (SVs)
- Satellites orbit the earth in around 12 hrs
- 6 orbital planes inclined at 55 degrees with the equator
- This constellation provides 5 to 8 SVs from any point on the earth
- Satellites carry station keeping fuel and maintained required orbits by occasional station keeping maneuvers



GPS Nominal Constellation
24 Satellites in 6 Orbital Planes
4 Satellites in each Plane
20,200 km Altitudes, 55 Degree Inclination

Introduction to GPS

Control Segment

- The **control segment** consist of **five monitor stations** and **four ground stations**
- **Monitor stations** measure the distances of the overhead satellites on **every 1.5 seconds** and **share** the corrected data to **Master control**
- Here the **satellite orbit**, **clock performance** and **health of the satellite** are determined and determines whether repositioning is required
- **Ground antennas** are also used to transmit commands to the Satellite and to receive the Satellite's sate of the art telemetry data

Introduction to GPS

User Segment

- It consists of **receivers** that **decode** the signals from the satellites
- The **receiver performs following tasks**:
  - Selecting one or more satellites
  - Acquiring GPS signals
  - Measuring and tracking
  - Recovering navigation data

Applications of GPS

- ❑ Timing atomic clocks
- ❑ Roads & Highways
- ❑ Space
- ❑ Aviation
- ❑ Agriculture
- ❑ Marine
- ❑ Railway
- ❑ Environment
- ❑ Public Safety & Disaster Relief
- ❑ Surveying & Mapping